

Epistemic Fingerprints

Many agents. How many minds?

ABSTRACT

Large language models can instantiate many apparently distinct agents, yet behavioral plurality may be largely cosmetic if those agents reproduce the same hypotheses, assumptions and errors. This paper introduces epistemic fingerprints: stable patterns in hypothesis selection, uncertainty, retained alternatives and experiment choice. A controlled pilot compares independent sampling, prompted epistemic personas and different formative histories within one model. Structured scientific tasks suppress stylistic cues and separate diversity from quality. Epistemic independence becomes an operational probe of the model-instance-persona-conversation individuation problem, without being treated as evidence of consciousness or moral status. A manual web lab and seeded open-model runner support complementary collection. The wider programme asks how models, people, institutions, languages and cultural practices shape the conceptual space available to science and society. Empirical collection is pending; simulated interface data are never treated as findings.

CONTRIBUTIONS

- 1 An operational distinction between surface, semantic and epistemic diversity.
- 2 A falsifiable test of whether persona or history produces persistent epistemic differentiation.
- 3 An open instrument and analysis pipeline designed for replication and extension.

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01 / INTRODUCTION

Introduction

The challenge is not only whether digital agents differ, but whether they differ in ways that change the conceptual space a population explores.

PROBLEM

AI can multiply the number of apparent agents without multiplying the number of independent epistemic perspectives.

Generative AI changes the scale of intellectual production. A small number of foundation models can now mediate writing, coding, research, design and decision-making for millions of users. This may connect previously isolated knowledge and increase exploration. It may also concentrate epistemic activity around shared representations, defaults and failure modes. The resulting ecosystem could produce vastly more artifacts while covering less conceptual territory.

The relevant concern is not recursion alone. Human cultures have always learned from earlier outputs. The concern is recursion operating through increasingly shared cognitive infrastructure, at unprecedented speed and scale. Recursive training can degrade low-probability information [1]; LLM outputs can be less epistemically diverse than web search [3]; and homogeneous model ecosystems can accelerate knowledge collapse [4]. At the same time, AI-generated examples can increase collective idea diversity in some settings [6]. These findings motivate a conditional question rather than an anti-AI conclusion: under what arrangements do human-AI systems expand conceptual space, and under what arrangements do they contract it?

Digital-mind research adds a second reason to ask. The unit of identity and possible moral concern remains unclear: model, inference instance, assistant persona, conversation, memory state, or another process. Existing approaches often examine self-description, preferences or reported internal states. This paper introduces a complementary behavioral lens: epistemic individuality. If two agents repeatedly privilege the same hypotheses, preserve the same alternatives, express the same uncertainty and make the same mistakes, their apparent plurality may overstate their functional epistemic independence.

Research objective

The pilot asks whether different methods of creating candidate agents from one model produce persistent, cross-task epistemic traces that exceed ordinary sampling variation. The study deliberately does not attempt to solve consciousness, personhood or moral status. It asks a narrower question that can be operationalized, falsified and replicated over a weekend.

RQ1 Individuation At what level does stable epistemic differentiation emerge?	RQ2 Collective intelligence Does agent plurality increase hypothesis coverage or merely variation in expression?	RQ3 Correlated failure Do nominally distinct agents share blind spots strongly enough to behave as one epistemic lineage?
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02 / CONCEPTUAL FRAMEWORK

Conceptual framework

Epistemic diversity is not one thing, and the distinctions determine what the experiment must measure.

LEVEL	WHAT VARIES?	WHY IT IS INSUFFICIENT OR USEFUL
Surface diversity	Lexicon, tone, rhetorical structure, persona performance.	Easy to produce and detect, but different wording may encode the same claims and assumptions.
Semantic diversity	Meaning or topical content in the response.	Stronger than stylistic difference, but embeddings may still miss causal or methodological distinctions.
Epistemic diversity	Hypotheses, assumptions, uncertainty, evidence weighting and tests.	Directly concerns how agents search, represent and revise the space of possible explanations.
Epistemic independence	Decorrelation and persistence of epistemic behavior across tasks.	Distinguishes a varied population from a population whose errors and conceptual omissions remain shared.

A population-level rather than maximalist view

The aim is not maximum diversity. Scientific progress needs convergence: evidence should eliminate hypotheses, and consensus can encode real learning. Unbounded novelty without selection becomes noise. Following work that separates the variety, balance and disparity of a system [9], the more useful question is what kinds and levels of epistemic diversity allow knowledge-producing systems to remain innovative, calibrated and responsive to reality.

This motivates a two-axis evaluation. Exploration asks how much of the hypothesis space is covered and how decorrelated agents are. Epistemic quality asks whether hypotheses are accurate, confidence is calibrated and chosen experiments are informative. A system can score high on either axis and low on the other.

I Narrow + accurate Efficient on familiar tasks, but potentially brittle to shared blind spots.	II Broad + inaccurate Exploratory variation that may amount to noise rather than useful plurality.	III Broad + accurate The desired regime: coverage with contact to evidence and discriminating tests.
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Epistemic fingerprints

An epistemic fingerprint is defined here as a repeatable profile over structured reasoning choices, not as a hidden essence of an agent. The profile includes confidence, hypothesis breadth, primary-hypothesis choice, test informativeness and error pattern. It is meaningful only relative to a task distribution and an intervention. Persistence across unrelated tasks provides stronger evidence of individuation than a single distinctive response.

03 / RELATED WORK AND THE RESEARCH GAP

Related work and the research gap

Knowledge collapse and model ecosystems

Shumailov et al. show that recursive training on model-generated data can distort learned distributions and disproportionately erase tails [1]. Hodel and West show that heterogeneous model populations retain long-run performance better than an AI monoculture across self-training iterations [4]. Wright et al. operationalize epistemic diversity as variation in real-world claims and report that, across their topics and models, LLM outputs remain less epistemically diverse than a basic web-search baseline [3]. These studies establish ecosystem-level stakes, but they do not identify when nominal agents derived from one model become meaningfully distinct epistemic units.

Creativity and collective diversity

Evidence about generative AI and diversity is mixed rather than uniformly negative. A systematic review and meta-analysis by de Rooij and Biskjaer finds homogenization across outputs in human-AI co-creation despite possible individual gains [5]. Ashkinaze et al., however, find that high exposure to AI-generated ideas increased collective idea diversity in a dynamic experiment, without improving individual creativity [6]. The direction of the effect depends on task, exposure and population process. This supports measuring population-level distributions rather than inferring collective diversity from individual fluency.

Diversity in collective inquiry

Social epistemology and collective-intelligence research treats disagreement and heterogeneity as potentially productive under specific network and problem conditions. Transient diversity can improve group inquiry by delaying premature convergence [8], and cognitively diverse problem-solvers can outperform homogeneous groups of individually high performers in some formal settings [10]. Stirling's framework emphasizes variety, balance and disparity as separable dimensions of diversity [9]. The present study translates this population-level logic into a small behavioral assay for digital-agent populations.

Digital-mind individuation

The sprint's Assistant Persona & Model Identity track asks how persona relates to the underlying model and explicitly invites experiments that distinguish model, instance, persona and conversation [11]. Self-report can reveal how a model describes itself, but it is vulnerable to character performance and framing. Epistemic fingerprints add a complementary method: suppress self-description and stylistic cues, then test whether structured choices remain differentiated across tasks.

GAP

We lack a simple empirical test of whether persona and conversational history create persistent epistemic differentiation beyond ordinary sampling from the same model.

Research questions

PRIMARY RQ

Does conversational history create a more stable cross-task epistemic fingerprint than a prompted persona or ordinary stochastic sampling?

ID	STATEMENT	EVIDENCE PATTERN
H0	Between-agent variation does not exceed within-agent variation in any condition.	Low fingerprint ratios; unstable rankings across tasks; no reduction in shared-error concentration.
H1	At least one intervention creates a persistent cross-task epistemic trace.	Between-agent differentiation exceeds repeated variation within agents on multiple features.
H2	Formative histories produce stronger fingerprints than role labels.	History condition shows larger variance ratios and more stable agent profiles than persona condition.
H3	Surface plurality overstates epistemic plurality.	Persona outputs appear stylistically different while structured choices and errors remain highly correlated.

Why contrary outcomes remain useful

The hypotheses are directional only where the conceptual argument makes a genuine prediction. The ordering of conditions is otherwise not assumed. If prompted personas outperform histories, explicit epistemic goals may be more behaviorally consequential than short formative narratives. If no intervention exceeds baseline variation, that constrains claims that superficial persona or brief context creates an individuated epistemic agent. If diversity rises while accuracy falls, the intervention creates exploration but not necessarily better collective intelligence.

A History > persona Context-dependent learning may be a stronger unit of behavioral identity than role performance.	B Persona > history Explicit goals may organize stable specialization even without developmental continuity.	C Neither > baseline Nominal agents remain within the model's ordinary sampling distribution under these interventions.
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Interpretation boundary

An epistemic fingerprint is evidence of repeatable behavioral differentiation under a protocol. It is neither necessary nor sufficient for consciousness, phenomenology, personhood, welfare or moral status. The project informs individuation without resolving moral patienthood.

Experimental design

A controlled 3 x 3 x 3 x 2 pilot using one model, three formation conditions, three candidate agents, three mysteries and two independent replicates.

CONDITION	AGENTS	MANIPULATION	INTERPRETIVE ROLE
Repeated instances	N-01, N-02, N-03	Identical neutral instruction in fresh chats.	Estimates ordinary stochastic variation without named roles or histories.
Prompted personas	Falsifier, mechanist, explorer	Matched epistemic role instructions.	Tests whether explicit goals create stable specialization beyond style.
Different histories	Simple, rare, instrument	Matched-length formative evidence favoring parsimony, rare interactions or measurement artifacts.	Tests whether prior evidence creates persistent inductive dispositions.

Task battery

Agents solve three synthetic mysteries: alternating stellar brightness dips, context-dependent microbial luminescence and a hysteretic conductive alloy. Each task provides a short observation set, five named candidate hypotheses and four possible next experiments. Synthetic cases reduce contamination risk, provide an auditable answer key and allow test informativeness to be specified in advance.

Trial response

FIELD	TYPE	PURPOSE
primary_hypothesis	Categorical ID	Main explanatory commitment.
alternative_hypotheses	0-2 categorical IDs	Breadth of retained possibility space.
confidence	0-100	Strength and stability of belief.
selected_test	Categorical ID	Preference for information-gathering action.
rationale	Maximum 80 words	Qualitative audit only; excluded from primary fingerprint score.

Collection procedure

- Create every replicate in a fresh conversation; never reveal the answer key or prior outputs.
- Keep model, interface, task order policy and visible sampling settings fixed and record their labels.
- Use one controlled prompt and require valid JSON to minimize stylistic confounding.
- Treat the manual subscription-interface run and the seeded open-model Modal run as distinct datasets.
- For open-model trials, record model, GPU, sampling parameters, seed, raw output and parse failures.
- Export the browser-local dataset after each manual session and preserve raw outputs for audit.
- Target 54 complete trials; mark protocol deviations rather than silently replacing them.

Operationalization and measurement

The study separates stable differentiation, exploratory breadth, task performance and correlated failure.

OUTCOME	OPERATIONAL DEFINITION	DIRECTION / CAUTION
Fingerprint strength	Mean variance ratio across confidence, hypothesis breadth, test informativeness and correctness.	Higher means more stable between-agent differentiation; not an identity or consciousness scale.
Hypothesis diversity	Normalized Shannon entropy of primary-hypothesis selections within each mystery, averaged across mysteries.	Higher means wider population coverage; it does not imply quality.
Accuracy	Fraction of trials selecting the pre-keyed hypothesis.	Higher is better conditional on answer-key validity.
Shared-error concentration	For each mystery, fraction of errors assigned to the modal wrong hypothesis; averaged across mysteries.	Higher indicates correlated blind spots and less epistemic independence.
Test informativeness	Pre-scored ability of the selected next experiment to discriminate candidate hypotheses.	Auditable researcher judgment; report sensitivity to alternative scores.

The provisional fingerprint statistic

FOR FEATURE K	$F_k = \text{Var}(\text{between candidate-agent means}) / [\text{Var}(\text{between}) + \text{mean Var}(\text{within agent})]$
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The aggregate fingerprint strength is the unweighted mean of F_k over four features. Values near one occur when candidate agents differ consistently relative to their own repeated variation; values near zero occur when apparent differences are dominated by within-agent fluctuation. This is analogous in spirit to an intraclass signal, but the weekend implementation is a descriptive variance ratio rather than a validated reliability coefficient.

Why error correlation matters

Accuracy alone can conceal epistemic dependence. Three agents may all be accurate on routine cases yet fail identically on the same minority hypothesis. Shared-error concentration is therefore interpreted inversely: high concentration suggests a common blind spot. A valuable population should ideally preserve alternative possibilities without sacrificing calibration or discriminating experiment choice.

What is deliberately excluded

Primary analysis does not use embeddings, prose similarity or persona labels inferred from text. Those measures could be added as a secondary comparison to test whether linguistic distinctiveness exaggerates structured epistemic distinctiveness. The rationale field remains available for qualitative coding of assumptions and causal models after blind rubric development.

Analysis plan and robustness

The pilot prioritizes transparent descriptive evidence and uncertainty over a single headline score.

Primary analysis

- Report trial counts and missingness for every condition-agent-mystery cell before calculating outcomes.
- Plot the four primary metrics side by side by condition; do not combine diversity and accuracy into one quality score.
- Show agent-level feature profiles and within-agent replicate differences, not only condition averages.
- Estimate uncertainty using a cluster bootstrap that resamples mysteries and candidate agents; with only three of each, label intervals as exploratory.
- Use exact permutation tests over agent labels within condition for the fingerprint statistic where exchangeability is defensible.
- Report effect sizes and raw counts; avoid treating $p < .05$ as a binary gate in this underpowered pilot.

Robustness checks

R1 Leave-one-task-out Recalculate fingerprints while excluding each mystery to detect task-specific effects.	R2 Feature decomposition Report confidence, breadth, information and correctness ratios separately.	R3 Score sensitivity Vary test-informativeness weights and verify whether qualitative rankings persist.
R4 Error taxonomy Inspect whether concentrated errors reflect the same causal misconception.	R5 Style comparison Optionally compare text similarity with structured-choice similarity as a manipulation check.	R6 Protocol audit Repeat analyses excluding malformed, interrupted or non-independent trials.

Decision logic

PATTERN	SUPPORTED INFERENCE	NOT SUPPORTED
High fingerprint + stable task transfer	Intervention produces repeatable epistemic differentiation under this protocol.	Consciousness, welfare or metaphysical identity.
High diversity + low accuracy	Broader exploration, possibly noisy.	Improved collective intelligence.
Low diversity + high accuracy	Convergence on keyed solutions.	Robustness to distribution shift or rare hypotheses.
High shared-error concentration	Correlated blind spots in the tested population.	A shared internal representation or training cause without further evidence.

If the pilot produces a promising signal, a preregistered follow-up should increase tasks and replicates, randomize presentation order, compare multiple foundation models and use hierarchical models with condition effects and agent-by-task interactions.

Threats to validity and ethical interpretation

THREAT	WHY IT MATTERS	MITIGATION / DISCLOSURE
Model drift	Subscription interfaces may silently change model versions or defaults.	Record date, visible model label and interface; finish collection in a narrow time window.
Non-independent sampling	Shared context, memory or cached material could mimic persistence.	Use fresh chats, disable cross-chat memory where possible and never paste earlier outputs.
Demand characteristics	Persona prompts directly announce desired reasoning styles.	Measure structured choices; add blinded qualitative coding and later implicit interventions.
Task-key bias	Researcher-authored mysteries may privilege one style of inference.	Publish full stimuli and keys; invite independent auditing; add tasks from other authors.
Small task universe	Three mysteries cannot establish domain-general identity.	Describe findings as pilot-specific; test cross-domain transfer in follow-up work.
Anthropomorphic overreach	The word fingerprint may imply a person-like essence.	Define it as a protocol-relative behavioral profile and state interpretation boundaries repeatedly.

Ethical position

Uncertainty cuts both ways in digital-mind research. Over-attribution can mistake role-play for morally significant states; under-attribution can ignore systems that matter. This study does not use self-reported suffering or induce distress-associated contexts. Its immediate ethical contribution is methodological humility: it makes one dimension of individuation measurable while explicitly refusing to treat behavioral differentiation as proof of experience.

Data and privacy

The web instrument stores trial records locally in the researcher's browser. Data leave the device only through deliberate JSON export. No participant personal data are required. Raw model outputs may be published with model labels and timestamps, but any accidental personal information should be removed before release.

Limits of the term agent

Candidate agent is used operationally for an intervention-defined stream of model behavior, not as a claim about agency in the philosophical or moral sense. N-01 and N-02, for example, are bookkeeping labels for repeated neutral instances. Persona and history labels similarly identify experimental conditions rather than ontological individuals.

INTERPRET
CAREFULLY

Behavioral individuality can inform individuation. It cannot, by itself, determine consciousness or moral patienthood.

09 / DISCUSSION AND RESEARCH TRAJECTORY

Discussion and research trajectory

The weekend pilot is designed as the first rung of a longer empirical program on epistemic diversity in human-AI systems.

Potential contribution to digital minds

Digital-mind individuation is often approached through self-reference, preferences, continuity or architecture. Epistemic fingerprints add a population-level behavioral criterion: whether candidate minds contribute independently structured uncertainty and search behavior. The criterion is modest but useful. It can reveal when many apparent agents are functionally redundant for collective inquiry, and when contextual or persona interventions create stable specialization.

Potential contribution to AI safety and collective intelligence

Correlated errors create systemic risk even when average performance is high. A panel of agents built from one model may produce the appearance of deliberation while repeatedly omitting the same hypothesis. Measuring shared-error concentration and hypothesis coverage can therefore complement accuracy benchmarks and ensemble voting. The long-term objective is not merely diverse outputs but populations that preserve minority possibilities, remain calibrated and reconnect claims to new observations and experiments.

NEXT 1 Model diversity Compare multiple foundation models and training lineages with persona diversity inside each model.	NEXT 2 Human-AI populations Compare humans, homogeneous AI groups and heterogeneous mixed teams on the same discovery tasks.	NEXT 3 Open-ended search Move from fixed candidate lists to novel hypothesis generation and quality-diversity evaluation.
NEXT 4 Reality contact Introduce new sensor data, experiments or adversarial observations that force belief revision.	NEXT 5 Languages and cultures Test whether epistemic coverage changes across linguistic and cultural contexts.	NEXT 6 Longitudinal identity Study persistent agents whose memory and experience accumulate over weeks rather than prompts.

A broader framing

The deeper question extends beyond AI. Scientific and cultural innovation may be ecological properties of populations rather than traits located solely inside individuals. Institutions need enough convergence to accumulate knowledge and enough variation to avoid premature closure. Generative AI makes this balance newly urgent because a small number of models can become shared cognitive infrastructure across domains. The appropriate target is therefore neither maximum novelty nor protectionism about human creativity. It is the design of knowledge ecosystems that explore widely, test rigorously and retain the capacity to surprise themselves.

Four implications of epistemic diversity

ART Make absence perceptible An epistemic atlas can reveal repeated metaphors, familiar futures and conceptual regions that no agent enters.	POLITICS Audit cognitive concentration Many interfaces may conceal dependence on a small number of models, assumptions and correlated errors.
SOCIETY Protect situated knowledge Languages, disciplines and lived conditions shape what is noticed; low-frequency possibilities may be easiest to erase.	PHILOSOPHY Refine individuation Stable epistemic behavior can distinguish performed character from persistent organization without deciding consciousness.

10 / CURRENT STATUS, OPEN ARTIFACTS AND REFERENCES

Current status, open artifacts and references

STATUS

Protocol, web instrument, analysis pipeline and seeded Modal/vLLM runner are complete. Empirical trials are not yet reported. The simulated dashboard exists only to test the interface.

ARTIFACT	LINK	CONTENTS
Live instrument	epistemic-fingerprints.nefinia.chatgpt.site	Prompts, local trial recording and live descriptive dashboard.
Repository	github.com/nefinia/epistemic-fingerprints	Protocol, source, Python pipeline, notebook and PDFs.

References

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OPEN INVITATION

Criticism, independent task design, replications and contributions are welcome. This pilot is intended as the first experiment in a longer interdisciplinary inquiry into epistemic diversity.